

RESEARCH INTERESTS

- Bayesian inference; uncertainty quantification; covariance estimation; multivariate data; conformal prediction; small area estimation; hierarchical modeling.

EMPLOYMENT

- **University of Chicago; Mansueto Institute for Urban Innovation** Chicago, IL
Mansueto Fellow; Advisors: Christopher Berry and James Evans September 2026 – present
- **MIT; Laboratory for Information & Decision Systems** Cambridge, MA
Postdoctoral Associate; Advisor: Tamara Broderick September 2024 – August 2026
- **Duke University** Durham, NC
Instructor of Record, STA 199: Introduction to Data Science and Statistical Thinking May 2024 - July 2024
- **NIEHS/NIH; Biostatistics and Computational Biology Branch** Durham, NC
Graduate Summer Intern; Advisors: Shanshan Zhao and Myeonggyun Lee June 2023 – July 2023
Project: Comparison of Frequentist and Bayesian Penalization Methods for Mixture Analyses
- **The Federal Reserve Board; Macroeconomics and Quantitative Studies** Washington, DC
Senior Research Assistant June 2017 – June 2019

EDUCATION

- **Duke University** Durham, NC
Statistical Science, PhD; Advisor: Peter Hoff Aug. 2019 – May 2024
Dissertation: Methodological Advances for Multi-group Data
NSF TRIPODs funding recipient Fall 2021
- **The University of Alabama** Tuscaloosa, AL
Applied Statistics, MS; Applied Statistics Masters Student of the Year (2017) Jan. 2016 – May 2017
Mathematics, BS with honors, minor in Economics Aug. 2013 – May 2017

PUBLICATIONS AND PREPRINTS

- ‘Feature Aware Covariance Estimation¹, With Application to Mixtures of Chemical Exposures’, **E. Bersson**, K. Hoffman, H. Stapleton, and D. Dunson, *Annals of Applied Statistics*, forthcoming.
- ‘The Curious Case of the Default Settings: Evaluating Default Performance of Variational Inference Software’, M. Anderson, **E. Bersson**, and T. Broderick, preprint, 2026.
- ‘Invited Book Review of ‘Robust Small Area Estimation: Methods, Theory, Applications, and Open Problems’, by Jiming Jiang and J. Sunil Rao’, **E. Bersson**, *Journal of the American Statistical Association*, 2026.
- ‘Contributed Discussion to ‘Sparse Bayesian Factor Analysis When the Number of Factors is Unknown’, by Frhwirth-Schnatter, S., Hosszejini, D., and Lopes, F. L.’, **E. Bersson**, *Bayesian Analysis*, 20(1): 322-324, 2025.
- ‘Frequentist Prediction Sets for Species Abundance Using Indirect Information’, **E. Bersson** and P.D. Hoff, *Journal of the Royal Statistical Society: Series A*, **with discussion**, 188(3): 658-673, 2025.
- ‘Bayesian Covariance Estimation for Multi-group Matrix-variate Data’, **E. Bersson** and P.D. Hoff, *Bayesian Analysis*, 20(4): 1481-1507, 2025.
- ‘Optimal Conformal Prediction for Small Areas’, **E. Bersson** and P.D. Hoff, *Journal of Survey Statistics and Methodology*, 12(5): 1464-1488, 2024. **2023 GSS/SRMS/SSS Student Paper Competition Winner**

¹Previously, ‘Covariance Meta Regression’.

- ‘*Bounded Rationality, Sticky Prices, and the Zero Lower Bound*’, **E. Bersson**, P. Hürtgen, and M. Paustian, *Journal of Money, Credit, and Banking*, 56(2-3): 365-393, 2023.

TALKS AND POSTERS

- Dr. Parmigiani’s Lab, Dana Farber Cancer Institute, Harvard University, 2025. ‘*Covariance Meta Regression*’, invited talk.
- Slovenian Month of Science, an initiative of the Slovenian government, 2025. ‘*Frequentist Prediction Sets for Species Abundance Using Indirect Information*’, invited talk.
- Department of Computer Science and Statistics, University of Rhode Island, 2025. ‘*Covariance Meta Regression*’, invited talk.
- BNP-14, 2025. ‘*Covariance Meta Regression*’, contributed talk.
- Department of Statistics, University of Massachusetts, Amherst, 2025. ‘*Covariance Meta Regression*’, invited talk.
- IDWSDS, 2024. ‘*Covariance Meta Regression*’, invited talk.
- Royal Statistical Society Annual Conference, 2024. ‘*Frequentist Prediction Sets for Species Abundance Using Indirect Information*’, with discussion, invited talk with travel support.
- ISBA Conference, 2024. ‘*Covariance Meta Regression*’, contributed talk with travel support.
- International Symposium on Nonparametric Statistics, 2024. ‘*Bayesian Covariance Estimation for Multi-group Matrix-variate Data*’, invited talk.
- The Bayesian Young Statisticians Meeting (BAYSM), 2023. ‘*Bayesian Covariance Estimation for Multi-group Matrix-variate Data*’, contributed talk.
- New Frontiers in Reliability and Risk Analysis Symposium: A Tribute to Nozer D. Singpurwalla, 2023. ‘*Bayesian Covariance Estimation for Multi-group Matrix-variate Data*’, contributed poster with travel support.
- Joint Statistical Meetings, 2023. ‘*Optimal Conformal Prediction for Small Areas*’, contributed talk with travel support.
- Notre Dame Future Faculty Workshop, 2023. ‘*Optimal Conformal Prediction for Small Areas*’, invited poster with travel support.
- ISBA Conference, 2022. ‘*Optimal Conformal Prediction for Small Areas*’, contributed poster with travel support.
- Consortium for the Advancement of Undergraduate Statistics Education, 2022. ‘*Supporting mentored undergraduate research in statistics*’, contributed talk with Joan Combs-Durso, Yue Jiang, and Edric Tam.

SERVICE

- Reviewer for WIDS-Cambridge poster session, 2026.
- Co-supervisor for a PhD student’s master’s thesis, MIT, 2024 – present.
- Reviewer for ASA SBSS Student Paper Competition, 2026.
- BAYSM scientific committee, 2025.
- Organizer of an invited session on citizen science data analysis for the RSS International Conference, 2025.
- Reviewer for the ASA SBSS Student Paper Competition, 2025.
- Co-organizer of Women in Statistics events, Department of Statistical Science, Duke University, 2022-2024.

- Graduate Consultative Committee, Department of Statistical Science, Duke University, 2022-2023.
- Mentor for the Undergraduate Thesis Proposal Writing Workshop, Duke University, May 2024.
- Project manager for the Mental Health and the Justice System group at Data+, Duke University, summer 2022.
- Graduate mentor for an independent study on Bayesian modeling, Duke University, spring 2022.
- Mentor for the Undergraduate Introduction to Research in Statistical Science Workshop, Duke University, May 2021.

TEACHING EXPERIENCE

- **Instructor** Duke University
STA 199: Introduction to Data Science and Statistical Thinking *Summer 2024*
- **Teaching Assistant**
Stat Inference, Regression Analysis (Head TA), Case Studies, *Duke University*
Statistical Consulting, Data Science (Head TA)
Discrete Mathematics, Applied Differential Equations II *The University of Alabama*
- **Funded invitations to future faculty workshops**
Preparing to Teach Statistics Workshop *University of Toronto Scarborough, 2023*
Future Faculty Workshop *University of Notre Dame 2023*